

Concrete Simulation Plan for a 48V vs 96V Humanoid Joint Powertrain Study

Overview

This document defines a concrete simulation plan for a thesis focused on higher-efficiency humanoid joint hardware, with emphasis on comparing 48V and 96V DC bus architectures, evaluating higher switching frequency and device technology, and quantifying motor, inverter, thermal, and runtime impacts.[cite:48][cite:52][cite:54][cite:64]

The study should be built around a single representative joint, ideally a knee or hip pitch joint, because a single well-instrumented actuator study is more defensible than a shallow full-robot study.[cite:48][cite:54] The central hypothesis is that a 96V architecture can reduce current-related losses and improve power density, but only if the motor and inverter are co-designed so that savings at the inverter are not offset by increased motor-side or thermal penalties.[cite:48][cite:52][cite:64]

Research Questions

The simulations should answer the following thesis questions:

1. How much efficiency improvement does a 96V joint DC link provide compared with 48V for the same mechanical output requirement?[cite:48][cite:52]
2. How do higher PWM switching frequencies change inverter losses, current ripple, motor copper loss, core loss, and thermal behavior?[cite:48][cite:52]
3. How much additional benefit is achieved when the motor is re-designed for 96V rather than simply reusing a 48V-oriented motor at a higher bus voltage?[cite:64][cite:48]
4. How do electrical and thermal improvements translate into higher continuous torque capability or longer robot runtime from a fixed battery energy budget?[cite:54][cite:56]

Simulation Objectives

The minimum simulation objectives should be:

- Build a validated 48V baseline joint model.[cite:17][cite:64]
- Compare 48V and 96V with the same motor to isolate pure bus-voltage effects.[cite:48][cite:52]
- Compare low and high switching frequencies with realistic device-loss models.[cite:48][cite:52]
- Develop a 96V-oriented motor model and compare it against the baseline motor in the same mechanical envelope.[cite:64][cite:115]
- Map all major losses into thermal rise and continuous operating capability.[cite:48][cite:51]

- Convert joint-level efficiency improvements into runtime or energy-per-task benefits.
[cite:54][cite:56]

Software Roles

PSIM

PSIM should be the primary environment for detailed inverter and power-stage simulations because it is well suited for switching-level studies, including deadtime behavior, device conduction loss, switching loss, gate-drive loss, DC link ripple, and frequency sweeps.[cite:48][cite:52]

PSIM should be used to generate detailed electrical loss maps over torque-speed operating points, switching frequency, and bus voltage. These maps should later be exported to higher-level models for runtime and thermal cycle evaluation.[cite:48][cite:52]

Simulink

Simulink should be used for the higher-level control and energy study rather than detailed transistor switching studies. It is best suited for long drive cycles, duty-cycle integration, simplified thermal RC simulation, and battery/runtime estimation once loss models have been identified in PSIM and motor losses have been characterized in ANSYS.[cite:54][cite:56]

Simulink should also be the place where MuJoCo-derived joint trajectories are used as realistic duty cycles, after the operating-point loss maps are validated.[cite:54]

ANSYS

ANSYS should be used for motor electromagnetic and thermal work. Maxwell should be used to derive motor parameters, torque-speed capability, back-EMF, resistance, inductance, copper loss, and core loss for both the 48V baseline motor and the 96V-oriented motor design.[cite:64][cite:115]

If available, Motor-CAD or equivalent ANSYS thermal tools should be used to map motor losses to winding, stator, and housing temperature rise under realistic cooling assumptions.[cite:48][cite:51]

Fusion 360

Fusion should be used for packaging, motor-envelope definition, housing layout, and basic mass and inertia extraction. It should not be the primary electrical simulation environment, but it is useful for ensuring the motor, reducer, PCB, and housing can realistically coexist in the actuator package.[cite:103][cite:117]

Recommended Study Structure

The thesis simulations should be organized in four layers.

Layer 1: Electrical Inverter and Bus Losses

This layer quantifies the effect of 48V vs 96V and switching frequency on:

- Phase RMS and peak current.[cite:48][cite:52]
- Bus current.[cite:48]
- MOSFET or GaN conduction loss.[cite:48][cite:52]
- Switching loss.[cite:48][cite:52]
- Gate-drive loss.[cite:48]
- Deadtime-related loss and waveform distortion.[cite:48]
- Cable and connector loss.[cite:8][cite:32]
- DC-link capacitor ripple current and voltage ripple.[cite:48][cite:52]

This layer should be implemented primarily in PSIM.[cite:48][cite:52]

Layer 2: Motor Electromagnetic Losses

This layer quantifies:

- Copper loss vs torque/current.[cite:64]
- Core loss vs electrical frequency and flux density.[cite:52][cite:64]
- Current-ripple-induced excess motor loss.[cite:52]
- Back-EMF margin and voltage utilization.[cite:48][cite:64]
- The effect of winding choice on 48V vs 96V operation.[cite:64][cite:115]

This layer should be implemented in ANSYS Maxwell, with loss tables exported to PSIM and Simulink.[cite:64][cite:115]

Layer 3: Thermal Behavior

This layer converts electrical and motor losses into temperatures and thermal limits. It should quantify:

- Switch junction temperature.[cite:48][cite:51]
- PCB or baseplate temperature.[cite:51]
- Motor winding temperature.[cite:48][cite:51]
- Housing temperature.[cite:48][cite:51]
- Continuous torque limit before thermal derating.[cite:48][cite:51]

This should begin with simplified RC thermal networks in PSIM or Simulink, then be checked against ANSYS thermal analysis for the final shortlisted designs.[cite:48][cite:51]

Layer 4: Energy and Runtime

This layer converts actuator efficiency into robot-level value. It should quantify:

- Energy per gait cycle.[cite:54][cite:56]
- Average joint electrical power.[cite:54][cite:56]
- Battery energy consumed per hour.[cite:54][cite:56]
- Runtime improvement for a fixed battery pack.[cite:54][cite:56]
- Optional pack-size reduction for equal mission duration.[cite:54]

This layer should be implemented in Simulink using drive cycles from MuJoCo only after operating-point maps have been established.[cite:54]

Joint and Drive-Cycle Definition

A single representative joint should be selected first, ideally knee or hip pitch. The model should include motor, inverter, reducer, reflected inertia, and a representative load trajectory.[cite:64][cite:103][cite:117]

Two classes of operating conditions should be used:

1. **Set-point operating points** for PSIM map generation, because detailed switching studies are easier to interpret when torque and speed are held constant over short windows.[cite:48][cite:52]
2. **MuJoCo-derived drive profiles** for higher-level Simulink energy and thermal-cycle studies, because they are more realistic but harder to use as a first-pass debugging environment for switching-loss identification.[cite:54]

Core Comparison Matrix

The simulation matrix should be fixed early so all later plots are consistent.

Variable	Cases
DC bus voltage	48V, 96V
Device technology	Silicon MOSFET baseline, GaN candidate [cite:48][cite:52]
PWM frequency	20 kHz, 40 kHz, 80 kHz, 100 kHz [cite:48][cite:52]
Motor design	Baseline 48V-oriented motor, 96V-oriented co-designed motor [cite:64][cite:115]
Thermal path	Baseline passive conduction, improved housing conduction
Operating condition	Fixed torque-speed map points; MuJoCo profile

Required Models

Model A: 48V Baseline Joint Powertrain

This is the reference model against which all improvements are measured. It should include:

- 48V DC source.[cite:17][cite:64]
- 3-phase inverter with realistic switch models.[cite:48]
- Baseline PMSM or BLDC joint motor from ANSYS-derived parameters.[cite:64][cite:115]
- Gear ratio and reflected inertia.[cite:103][cite:117]
- Current control and speed or torque control loop.[cite:48]
- Fixed operating points and later a representative gait profile.[cite:54]

Model B: 96V Bus with Baseline Motor

This model changes only the DC bus and appropriate device class. It isolates the benefit of 96V without motor redesign.[cite:48][cite:52]

This model should answer whether 96V alone lowers current, cable losses, and inverter conduction losses enough to justify the added device and safety complexity.[cite:48][cite:52][cite:45]

Model C: 96V Bus with Co-Designed Motor

This model uses the 96V bus and a revised motor design with winding and electrical parameters better matched to the higher voltage and switching strategy.[cite:64][cite:115]

This is the key research model because it shows whether motor-inverter co-design produces net gains beyond the simple bus-voltage increase.[cite:48][cite:64]

Recommended Operating-Point Map

PSIM should first use a grid of fixed operating points rather than long drive profiles. A suitable first map is:

- Speed: 10%, 30%, 50%, 70%, 90% of max joint speed.
- Torque: 10%, 25%, 50%, 75%, 100% of continuous torque.
- Optional overload line: 125% and 150% torque for short-duration thermal checks.

At each point, record:

- DC input power.
- Mechanical output power.
- Efficiency.
- RMS phase current.
- Peak phase current.

- Inverter conduction loss.
- Inverter switching loss.
- Gate-drive loss.
- Motor copper loss.
- Motor core loss.
- Total system loss.

Detailed PSIM Tasks

The following PSIM tasks should be completed in order.

Task 1: Device-Level Baseline

Build 48V and 96V inverter models with realistic device data. Include deadtime, gate-drive voltage, switching transitions, and thermal impedance where available.[cite:48][cite:52]

Outputs: switching waveforms, phase current waveforms, deadtime-distortion examples, switching loss per device, gate-drive loss, and bus ripple plots.[cite:48][cite:52]

Task 2: Frequency Sweep

For each bus voltage and device technology, sweep switching frequency from 20 kHz to 100 kHz across selected torque-speed points.[cite:48][cite:52]

Outputs: total inverter loss vs switching frequency, motor loss vs switching frequency, and total efficiency vs switching frequency.[cite:48][cite:52]

Task 3: Bus-Voltage Comparison

At equal mechanical output, compare 48V and 96V using the same motor. Quantify current reduction, cable loss reduction, inverter conduction-loss change, and total efficiency shift.[cite:48][cite:52]

Outputs: side-by-side loss breakdown and current comparison plots.[cite:48][cite:52]

Task 4: Co-Designed 96V Motor Comparison

Replace the baseline motor with the 96V-oriented motor and repeat the key operating points and sweeps.[cite:64][cite:115]

Outputs: comparison of baseline 48V motor vs 96V-oriented motor under the same package constraints, including motor loss split and total actuator efficiency.[cite:64][cite:115]

Detailed ANSYS Tasks

Task 1: Baseline Motor Extraction

Create a baseline motor model sized for the chosen joint envelope and 48V design assumptions. Extract:

- Torque constant and back-EMF constant.[cite:64]
- Resistance and inductance.[cite:64]
- Copper loss vs current.[cite:64]
- Core loss vs speed/frequency.[cite:52][cite:64]
- Torque-speed envelope.[cite:64]

Task 2: 96V-Oriented Motor Design

Modify the motor design so that it fits the same package but is electrically better matched to a 96V bus and the target switching strategy. This likely involves winding and conductor tradeoffs rather than a completely different actuator form.[cite:64][cite:115]

Extract the same quantities as above so fair comparisons can be made.[cite:64][cite:115]

Task 3: Thermal Mapping

Run thermal cases for representative losses to determine winding and housing temperature under passive conduction assumptions. Use at least three points:

- Light-load continuous case.
- Nominal walking case.
- High-torque burst or near-continuous limit case.[cite:48][cite:51]

Detailed Simulink Tasks

Task 1: Duty-Cycle Energy Model

Use MuJoCo-derived joint torque-speed trajectories as the primary duty-cycle input for the energy model, but not for first-pass switching-loss debugging.[cite:54]

Implement a joint power block where instantaneous losses are obtained from lookup tables generated from PSIM and ANSYS results.[cite:48][cite:54][cite:64]

Task 2: Battery and Runtime Model

Model a fixed battery pack and compute:

- Energy consumed per cycle.[cite:54][cite:56]
- Energy consumed per hour.[cite:54][cite:56]

- Runtime before depletion.[cite:54][cite:56]
- Optional improvement in runtime or required pack size.[cite:54]

Task 3: Simplified Thermal Cycle Model

Use lumped thermal RC networks in Simulink to estimate transient temperatures over long task profiles. This should be driven by power-loss time histories derived from the duty-cycle simulation.[cite:48][cite:51]

Required Thesis Figures

The thesis should include the following graphical outputs.

Electrical and Loss Figures

1. Block diagram of the simulation workflow showing PSIM, ANSYS, Simulink, and data exchange paths.
2. Joint actuator architecture diagram showing battery, DC bus, inverter, motor, reducer, and thermal paths.
3. Representative switching waveforms at one operating point for 48V and 96V, including phase voltage and current.[cite:48][cite:52]
4. Deadtime effect plot showing current distortion or diode-conduction interval before and after deadtime optimization.[cite:48]
5. Loss breakdown bar charts at selected operating points, split into cable, inverter conduction, inverter switching, gate-drive, copper, and core losses.
6. Efficiency map over torque-speed space for 48V baseline.[cite:48][cite:64]
7. Efficiency map over torque-speed space for 96V baseline motor.[cite:48][cite:52]
8. Efficiency map over torque-speed space for 96V co-designed motor.[cite:48][cite:64]
9. Current comparison plot showing RMS and peak current at equal mechanical output for 48V vs 96V.
10. Switching-frequency sweep plot showing total actuator loss vs frequency for 48V and 96V.[cite:48][cite:52]
11. Motor-loss split plot showing copper vs core vs ripple-related loss across frequency.[cite:52][cite:64]

Thermal Figures

12. Thermal RC network diagram for inverter and motor.
13. Steady-state temperature plot for junction, winding, and housing across operating points.[cite:48][cite:51]
14. Temperature rise vs time for a representative gait or repeated duty cycle.[cite:48][cite:51]
15. Continuous torque capability plot before hitting thermal limit for 48V vs 96V architectures.[cite:48][cite:51]

Runtime and System Figures

16. Drive-cycle power trace from MuJoCo-derived profile.[cite:54]
17. Instantaneous loss trace over the same cycle.
18. Energy per cycle comparison for 48V, 96V without motor redesign, and 96V with co-designed motor.[cite:54][cite:56]
19. Runtime comparison chart for a fixed battery pack.[cite:54][cite:56]
20. Sensitivity plot showing how gains change with switching frequency, cooling quality, or cable length.

Required Thesis Tables

The thesis should include at least the following tables.

Table	Purpose
Joint design assumptions	Joint type, gear ratio, speed, torque, inertia, duty cycle
Device comparison	Si vs GaN device ratings, Rds(on), switching energy assumptions [cite:48][cite:52]
Motor parameter comparison	Baseline motor vs 96V-oriented motor [cite:64][cite:115]
Thermal assumptions	Ambient, interface resistance, housing conduction assumptions
Battery/runtime assumptions	Pack voltage, usable energy, auxiliary loads [cite:54][cite:56]

Minimum Publishable Result Set

If scope becomes constrained, the minimum thesis-quality result set should still include:

- One representative joint.[cite:64]
- 48V baseline, 96V baseline-motor case, and 96V co-designed-motor case.[cite:48][cite:64]
- At least one frequency sweep series.[cite:48][cite:52]
- At least one thermal comparison.[cite:48][cite:51]
- At least one runtime or energy-per-cycle comparison using a realistic task profile.[cite:54][cite:56]

This minimum set is sufficient to support a defensible claim about whether 96V and co-design produce system-level benefits for humanoid joint hardware.[cite:48][cite:54][cite:64]

Recommended Chapter or Paper Narrative

A clean thesis or paper structure would be:

1. Motivation: humanoid battery and efficiency constraints.[cite:54][cite:56]
2. Hypothesis: 96V reduces current-related losses, but co-design is required for full benefit.[cite:48][cite:64]
3. Method: PSIM + ANSYS + Simulink workflow.

4. Baseline: 48V actuator performance.[cite:17][cite:64]
5. Electrical comparison: 48V vs 96V with same motor.[cite:48][cite:52]
6. Co-design comparison: 96V baseline motor vs 96V-oriented motor.[cite:64][cite:115]
7. Thermal and continuous capability results.[cite:48][cite:51]
8. Runtime and system-value implications.[cite:54][cite:56]
9. Design guidelines for future humanoid joint architectures.[cite:48][cite:64]

First Four Weeks Execution Plan

Week 1

- Freeze joint selection, gear ratio, continuous and peak torque, and speed range.
- Define the fixed operating-point map and select MuJoCo trajectory segments.
- Create the initial Fusion package envelope.

Week 2

- Build 48V baseline motor in ANSYS.
- Extract motor electrical parameters and first-pass loss tables.
- Build initial 48V inverter in PSIM.

Week 3

- Run 48V baseline operating-point simulations.
- Add deadtime and realistic device-loss modeling.
- Generate first loss breakdown plots.

Week 4

- Move to 96V with baseline motor in PSIM.
- Run frequency sweep and bus-voltage comparison.
- Identify what must change in the motor design for the 96V-oriented case.

Final Guidance

The most important methodological choice is to separate **operating-point identification** from **realistic duty-cycle integration**. PSIM should first be used with fixed torque-speed set points to create trustworthy loss maps, and only then should MuJoCo drive profiles be used in Simulink to turn those maps into thermal and runtime results.[cite:48][cite:52][cite:54]

The most important contribution opportunity is not merely showing that 96V lowers current. The stronger contribution is demonstrating whether a 96V architecture, combined with higher-frequency inverter design and a motor re-optimized for that bus, produces a net benefit in

efficiency, thermal headroom, and usable runtime for humanoid joints.[cite:48][cite:52][cite:54]
[cite:64]